

Lab Task 12

University Student Portal — Authentication System

Android Studio • Kotlin • Firebase Authentication



Instructions

1. Complete the task following all requirements carefully.
2. After completion, push your project to your GitHub repository.
3. UI must be clean, aligned, and properly spaced.
4. Test all authentication flows: register, login, logout, and password reset.
5. Firebase project must be properly connected — verify google-services.json is present.



Scenario

You are building an authentication system for a University Student Portal. Students can register with their email and password, log in securely, reset forgotten passwords via email, and log out. The app uses Firebase Authentication to manage all auth flows. After login, students are directed to a home dashboard displaying their profile info. Unauthenticated users cannot access the dashboard.



Project Setup

Project Name	StudentAuthApp
Template	Empty View Activity
Language	Kotlin
Firebase Service	Firebase Authentication (Email/Password)

Step 1 — Add Firebase to Your Project

- Go to Firebase Console (console.firebase.google.com) → Create a new project
- Add an Android app — enter your package name (e.g. com.university.studentauth)
- Download google-services.json and place it in the app/ directory
- Add the Firebase BOM and Authentication dependency to build.gradle:

Step 2 — Enable Email/Password Sign-In

- In Firebase Console → Authentication → Sign-in method → Enable Email/Password
- Save the change

✓ Requirements & Screens

Screen 1 — Splash / Launcher Screen (SplashActivity)

Layout: ConstraintLayout

- Display university logo (ImageView) and app name centered on screen
- After a 2-second delay, check authentication state:
 - If user is already logged in → navigate to HomeActivity (skip login screen)
 - If not logged in → navigate to LoginActivity
- Use FirebaseAuth.getInstance().currentUser != null to check login state

Screen 2 — Register Screen (RegisterActivity)

Layout: ConstraintLayout + ScrollView

- Full Name — EditText (TextPersonName input type)
- Email — EditText (TextEmailAddress input type)
- Password — EditText (TextPassword input type, toggle visibility icon)
- Confirm Password — EditText (TextPassword input type)
- Register button — creates account via Firebase
- "Already have an account? Login" — TextView link to LoginActivity
- Validation rules before calling Firebase:
 - All fields must be non-empty
 - Email must be a valid format (use Patterns.EMAIL_ADDRESS.matcher())
 - Password must be at least 8 characters
 - Confirm Password must match Password
- On success → navigate to HomeActivity, finish RegisterActivity
- On failure → display Firebase error message in a Snackbar
- Show ProgressBar while registration is in progress

Screen 3 — Login Screen (LoginActivity)

Layout: ConstraintLayout

- Email — EditText
- Password — EditText (with toggle visibility)
- Login button — signs in via Firebase Email/Password
- "Forgot Password?" — TextView link to ForgotPasswordActivity
- "Don't have an account? Register" — TextView link to RegisterActivity
- Show ProgressBar while login is in progress
- On success → navigate to HomeActivity, finish LoginActivity (no back to login)
- On failure → show Snackbar with Firebase error (e.g., wrong password, user not found)

Screen 4 — Forgot Password Screen (ForgotPasswordActivity)

Layout: ConstraintLayout

- Email — EditText
- Send Reset Email button — calls `auth.sendPasswordResetEmail(email)`
- On success → show Toast "Reset email sent. Check your inbox." and finish activity
- On failure → show error message in Snackbar
- Back arrow to return to LoginActivity

Screen 5 — Home Dashboard (HomeActivity)

Layout: ConstraintLayout

- Welcome card showing: user's email, UID (first 8 chars), account creation date
- "You are logged in" status badge (green indicator)
- Logout button — calls `auth.signOut()`, navigates to LoginActivity, clears back stack
- Change Password section: two EditTexts (New Password, Confirm Password) + Update button
 - Use `currentUser?.updatePassword(newPassword)` to update
- Delete Account button — shows a confirmation AlertDialog before calling `currentUser?.delete()`
- Profile initial avatar — a CircleView or simple TextView styled as a circle showing first letter of email